VIRTULIZATION

Hardware	Network	Storage	Memory	Software	Data	Software

• Full
  • Bare-Metal
  • Hosted
• Partial
• Para
	• Internal Network Virtulizaion
	• External Network Virtulization
		• Block Virtulization
		• File Virtulization
			• Application Level
			• OS Level Integration
				• OS Level
				• Application
				• Service
					• Database	• Virtual desktop Infrastructure
						• Hosted Virtual Desktop

Figure 1: Types of virtualisation

files, system libraries and configuration files, applications, ports, and routing rules. OpenVZ is an open source product available under the GNU GPL (General Public License).

So how do OpenVZ containers differ from the traditional virtual machine architecture? Well, they run on the same OS kernel as the host system, but allow multiple Linux variants in individual containers. This single-kernel implementation enables running containers with much less overhead. Hence, OpenVZ offers higher efficiency and manageability than traditional virtualisation technologies. It uses a single patched Linux kernel, and as a result can run only Linux—and doesn't have the overhead of a hypervisor (a tiny part of the CPU resources is used on virtualisation—around 1-2 per cent); it is fast and efficient.

OpenVZ features

The main features are listed below.

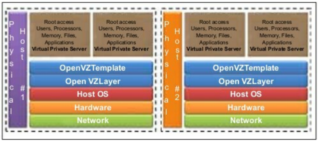

Figure 2: OpenVZ architecture

- OpenVZ uses a single kernel (Linux) implementation and hence it is as scalable as the Linux kernel.
- Virtualisation overhead is very low (approximately 1-2 per cent).
- Live migration of VPS and the checkpointing feature allows users to migrate a VPS from one physical host to another without needing to shut down the VPS.
- Resource management allows OpenVZ to share available host system resources among VPSs in an efficient manner; it guarantees QoS—not only providing performance, resource isolation, and protection from denial-of-service attacks, but also collecting usage information to monitor the system's health.
- By default, direct access to hardware is not available.
- OpenVZ has undergone a thorough security audit, which was performed by Solar Designer.
- IPsec is supported inside containers since kernel version 2.6.32.
- OpenVZ technology scales up to thousands of CPUs and terabytes of RAM.

Therefore, the benefits include near-zero overhead, strong isolation, improved flexibility, efficiency, and quality of service. Note that Oracle, DB/2, WebLogic, WebSphere and other big applications run efficiently inside OpenVZ containers. Applications and services need not be OpenVZ-aware. Along with standardised server management, note that OpenVZ kernels are based on Red Hat Enterprise Linux kernels, which are conventional and well maintained. By default, OpenVZ restricts container access to physical devices, making containers hardware-independent.

Its limitations are that OpenVZ supports only Linux distributions and not Windows. The */dev/loopN* devices are often restricted in deployments, which restricts the ability to mount disk images. OpenVZ is restricted to providing only a few VPN technologies based on Point-to-Point Protocol (PPP), such as PPTP Point-to-Point Tunnelling Protocol (PPTP), Layer 2 Tunnelling Protocol (L2TP) and TUN/TAP (virtual network kernel devices).

OpenVZ Use-Cases

Development and testing	Developers and testers need access to different Linux variants to develop and test an application—hence, testing and development groups often require a lot of hardware resources. By using OpenVZ, developers and testers can create multiple partitions with different Linux variants and configurations residing on one physical host.
Hosting	Useful to have isolated users and cost-efficient containers that behave like a server, to support multiple versions of an application, and offer easy administration.

A comparison of OpenVZ with its counterparts

How does OpenVZ compare with other virtualisation technologies such as Xen, KVM, VirtualBox, VMware Player and VMware Workstation? Table 1 gives a comparison.